1) Assumption Mapping & Design Space

Purpose

This document captures the **design space before commitment**.
You are not choosing a solution yet.

B. Core System Functions

List **what the system must do**, not how. (add all core functions)

Function ID	Function Description
F1	Capture clear visual information of the user's pupils during inspection tasks
F2	Remain comfortable and usable for inspectors during extended wear
F3	Physically integrate sensors/camera without blocking the forward field of view
F4	Operate reliably under industrial lighting and reflective surfaces
F5	Maintain stable relative geometry between eyes and sensors during head and body movement
F6	Establishing a reliable communication link between the embedded processor and camera modules (eye camera + scene camera), negotiating formats, and acquiring raw image data
F7	Controlling and sustaining stable capture parameters (resolution, FPS, pixel format) under varying thermal and computational load conditions
F8	Assigning each acquired frame a precise, monotonically increasing timestamp and maintaining temporal coherence between dual camera streams
F9	Decoupling frame acquisition from downstream processing/storage to absorb transient latency spikes and prevent frame loss
F10	Compressing and persisting video streams and structured metadata to onboard storage in a format suitable for later analysis
F11	Moving recorded sessions from the embedded device to an external system for post-processing, analysis, and archival
F12	Localize the pupil center in each eye image frame

F13	Track pupil position across consecutive frames
F14	Estimate gaze direction using calibration mapping
F15	Maintain stable gaze estimation under moderate head motion
F16	Transform pupil coordinates into world (inspection scene) gaze coordinates
F17	Detect and quantify inspection coverage using gaze heatmaps
F18	Detect deviation from predefined SOP inspection zones
F19	Store, compress, and manage inspection session data efficiently
F20	Quantitatively measure gaze accuracy
F21	Define temporal resolution and the latency evaluation framework
F22	Define comparative benchmarking framework (Kexxu available)
F23	Analyze system robustness under environmental variation
F24	Define the cost-performance evaluation model
F25	Define the failure mode characterization process
F26	Structure results for stakeholder verification and reporting

C. Design Space Exploration (Concept Fragments)

For each function, list **fundamentally different mechanisms**. You are encouraged to deliberate on radically different design concepts (don't hold back)

Function	Mechanism A	Mechanism B	Mechanism C
F1	Two inward-facing cameras, mounted near each eye	Single central camera with mirrors.	IR photodiode arrays
F2	Glasses style frame with nose pads	Helmet or head-mounted frame	Clip-on module attached to safety glasses

F3	Thin camera modules embedded in the frame rim	External camera pods attached to the frame.	Fibrescope (fibre optic camera)
F4	Infrared illumination with bandpass filters	Visible-light cameras with polarizing filters	Shielded optical tunnel to block ambient light
F5	Rigid glasses frame with fixed mounts	Adjustable articulated camera mounts	Headband-mounted optical module, separate from the frame
F6	V4L2(Video4Linux2) Direct Device Access	GStreamer	OpenCV VideoCapture
F7	Fixed Negotiation (Static Configuration)	Adaptive Scaling (Dynamic Degradation)	Multi-Profile Capture
F8	Software Monotonic Clock Stamping	UVC Payload Header Hardware Metadata	Hardware Trigger Synchronization (External Sync Signal)
F9	Per-Camera Ring Buffer (Bounded Queue)	Unbounded Queue with Backpressure Signaling	V4L2 Kernel-Level Multi-Buffer (Zero-Copy DMA)
F10	MJPEG Pass-Through (Camera-Side Compression)	Hardware H.264/H.265 Encoding (SoC Encoder)	Lossless Frame Archival (Raw/PNG per Frame)
F11	Physical Wired Transfer (USB/Ethernet File Copy)	WiFi-Based File Transfer / Streaming	Bluetooth Low Energy (BLE) for Metadata / Control
F12	Adaptive image segmentation with contour extraction	Edge detection with ellipse fitting	ML-based pupil detection model
F13	Independent frame-by-frame detection	Kalman filter prediction model	Optical flow-based motion tracking
F14	Linear geometric calibration	Polynomial regression calibration	Learned a regression mapping model

F15	Restricted head motion assumption	Geometric compensation model	Dual-eye fusion stabilization
F16	Polynomial regression mapping	Homography transformation	Perspective geometric model with calibration grid
F17	2D Gaussian accumulation	Grid-based zone counting	Fixation-duration weighted mapping
F18	Rule-based zone thresholding	Sequence validation logic	Time-based minimum dwell enforcement
F19	Store full video + gaze overla	Store gaze coordinates only	Hybrid (video for failed cases only)
F20	The subject looks at digital points on a monitor to measure pure angular error.	The subject walks a physical space, touching fixed markers to test spatial mapping accuracy.	Feed the system pre-rendered, 3D eye videos to test pure software accuracy.
F21	Measure Visual lag by recording the user's eye and the system's output screen.	Perform an action (e.g., flash LED), measure the ms delay in registering.	Analyze the delta between frame-capture timestamps and output-generation timestamps in the logs.
F22	Mount our design with Kexxo, and capture identical data simultaneously.	User performs a task with Kexxo, and then our prototype.	Feed both Kexxo and our prototype pre-recorded videos and compare the output.
F23	Controlled lab lighting variation.	In-Situ Factory Testing	Digital Noise Addition (Stress Test)
F24	Hardware (Bill of Materials) comparison	Total Cost Ownership (monetary + time)	Defect Reduction ROI
F25	Automated Boundary Testing: feed the algorithm extreme gaze angles until tracking drops.	Intentionally sabotage tests to map failure states.	Reverse-engineer the cause from data captured during natural failures.
F26	Standardized Academic Paper	Interactive Digital Dashboard	Executive Summary Matrix, mapping to Toyota's initial requirements

D. Assumption Mapping

List assumptions that **must be true** for a design to succeed.

Assumption ID	Assumption	Why It Matters	What Breaks if False
A1	Factory lighting remains within the expected intensity range	Affects pupil contrast and glare	Eye images become unusable due to saturation or low contrast
A2	Reflections from car surfaces do not dominate camera input	Reflections can confuse pupil detection	Tracking accuracy collapses
A3	Inspectors will wear the glasses correctly and consistently	Geometry depends on placement	Invalid calibration results in the wrong gaze mapping
A4	Users tolerate additional weight on the face	Comfort affects adoption	The device is not used or used incorrectly
A5	Inspector's eye shapes fall within the expected anatomical range	Optics are designed for average geometry.	Invalid calibration, uncomfortable frame, resulting in the device not being used or getting damaged.
A6	Sufficient Local Storage	Allows uninterrupted recording for analysis sessions	Recording stops early, or data loss occurs
A7	Dual UVC Cameras	Defines system architecture and simplifies integration using standard V4L2 drivers.	Multi-device sync becomes complex; pipeline design must change.
A8	Scene Recording: 1080p @ 30fps (MJPEG)	Sets expected data rate and performance baseline for scene capture	Mismatch affects storage, bandwidth planning, and sync accuracy.
A9	Eye Recording: 1280x800 @	Higher FPS is critical for	Lower FPS reduces gaze precision and

	60fps (MJPEG)	accurate eye movement tracking.	temporal resolution.
A10	Standard uvcvideo Driver	Ensures compatibility with V4L2 and standard Linux tools.	Custom drivers or incompatibility increase development complexity.
A111	USB Bandwidth Headroom	Ensures stable simultaneous capture without frame drops.	Frame drops, reduced FPS, or camera failure under load.
A12	OS: Ubuntu / Yocto	Defines software stack and toolchain compatibility.	Porting issues and inconsistent environment behavior
A13	Soft Real-Time Pipeline	Allows minor delays without system failure.	Strict real-time constraints would require a different design.
A14	Offline Processing First	Simplifies pipeline by avoiding immediate real-time constraints.	Real-time requirements would need major architectural changes.
A15	Output: MP4 + JSONL	Ensures compatibility with analysis tools and workflows.	Data becomes harder to process or integrate
A16	The operator maintains moderate head stability	Calibration assumes limited movement	Gaze error increases
A17	Frame rate remains stable (~60 FPS)	Tracking relies on temporal continuity	Tracking becomes noisy
A18	Pupil remains approximately circular in the image	Ellipse fitting assumes circular geometry	Detection accuracy drops
A19	Calibration mapping remains stable during inspection	Mapping accuracy depends on geometry	Gaze overlay becomes inaccurate
A20	Gaze corresponds to inspection	Compliance logic assumes visual focus equals	False compliance results

	attention	inspection	
A21	Frame timestamps are reliable	Synchronization accuracy required	Incorrect gaze-scene alignment
A22	SOP inspection zones are clearly defined	Analytics depends on defined ROIs	Deviation logic invalid
A23	The lighting and conditions we test in are similar to what actually happens on the Toyota inspection floor.	If we only test in perfect lab lighting, the system might look accurate, but real factory lighting has glare, shadows, and reflections.	The system might pass our lab tests but fail when used in the real factory.
A24	The Kexxu glasses are accurate enough to be used as a fair comparison device.	We are comparing our system's performance against Kexxu.	If Kexxu's data is inherently flawed or drifting, we will be tuning our system to match incorrect data.
A25	The system responds fast enough so that the gaze marker keeps up with where the person is actually looking.	If the system is slow, the gaze point may appear slightly behind the user's real eye movement.	The system might look inaccurate during fast movements, even if the eye detection itself is correct.
A26	The physical testing apparatus and procedures do not alter the inspector's natural factory behavior.	System must be validated on real-world movements and gaze patterns.	If the testing setup isn't good, our validation won't translate to factory success.
A27	Toyota's definition of a "successful inspection" can be translated into quantifiable data points.	Defining a cost-performance evaluation model (F5), and other verifications require hard numbers	We won't be able to prove system "success" if the criteria cannot be measured.

For F1:

Assumption ↓ / Mechanism →	Dual Cameras	Central Camera + Mirrors	Photodiodes
Lighting is stable	✓	✓	x
User wears correctly	✓	✓	✓
Reflection manageable	✓	x	✓
High accuracy needed	✓	✓	x

E. Known Unknowns

Unknown	Why It's Uncertain	Potential Impact
Actual glare level from painted car bodies	Depends on the factory environment	May blind sensors
Long-term comfort over full work shifts	Not yet tested	User rejection risk
Optimal camera angle relative to the eye	Depends on the algorithm tolerance	Poor accuracy if misaligned
Effect of sweat, dust, and oil vapors	Environment specific	Optical contamination
Calibration time acceptable to workers	Depends on the worker's workflow	Operational rejection
Does the UVC metadata stream contain valid PTS/SCR timestamps?	The /dev/video3 metadata node exists, but the actual content of the UVCH payload headers has not been validated	If the PTS/SCR fields are zero or unsupported, hardware-based timestamp synchronization is unavailable, and software timestamps become the only option
Actual inter-camera timestamp drift over extended sessions (10–60 min)	Software timestamps from two independent capture threads on different USB devices will accumulate clock skew	If drift exceeds the interframe interval (~16.7 ms at 60 fps), temporal alignment becomes ambiguous, and gaze-to-world mapping degrades. This would force

		the evaluation of a hardware trigger or metadata-based alternative
Does the embedded board thermally throttle under sustained dual-camera capture?	Continuous video processing generates significant heat. Throttling reduces CPU frequency and can cascade into frame drops	If thermal throttling occurs within the first 10 minutes, all mechanisms assuming stable FPS throughout a session are invalidated. Active cooling or dynamic degradation would become mandatory
Actual MJPEG decode overhead when re-encoding to H.264	The camera's output is MJPEG; H.264 encoding requires first decoding MJPEG to raw, then encoding to H.264	If MJPEG decode + H.264 encode exceeds ~10% CPU, the pipeline may not sustain 60 fps without hardware encoder support. This would force either MJPEG pass-through storage or reducing FPS
Detection accuracy under factory glare	Real lighting not fully tested	Reduced reliability
Long-term calibration drift	Depends on operator behavior	Increasing gaze error
Performance difference between classical CV and ML	Depends on training data	May require redesign
Acceptable gaze error tolerance	Integration requirements unclear	Need for system tuning
Real-time performance under load	Processing power constraints unknown	Latency issues
Acceptable gaze error margin (in pixels)	Depends on inspection zone size	Misclassification risk
Natural gaze jitter magnitude	Worker dependent	Noisy heatmaps
Acceptable dwell time threshold	Industry-specific	Over/under deviation detection
Long-duration calibration stability	Not yet tested	Drift over time
Physical Interoperability	It is unknown if we can physically mount our custom prototype alongside the Kexxu glasses simultaneously.	This limits our comparative benchmarking to sequential testing (testing one device after the other), which introduces human behavioral variance into the data.
Primary Performance Metric Weighting	Toyota's primary operational complaint with Kexxu is *connectivity,* but also gaze-plots, visual focus maps.	It is undefined what we will weigh in the final performance score.
Initial Lab Testing Apparatus	Since we have not yet conducted a site visit to the Toyota factory, the exact physical setup needed for our initial lab	Our initial lab benchmarks might fail to replicate the specific environmental edge cases of the inspection line.

	testing is undefined.	
Availability of Secondary Ground Truths	Currently, Kexxu is our only baseline reference. It is unknown if we can secure secondary validation hardware if Kexxu proves inadequate for specific edge-case comparisons.	If the Kexxu system fails in certain conditions, we will completely lack a comparative baseline for validation.